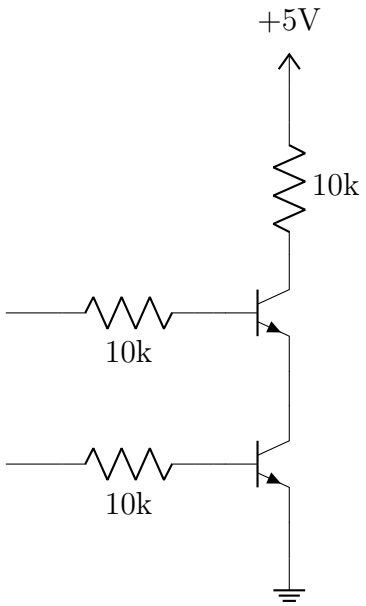
Questions

Question[10]: Identify the circuit, write its truth table, and electrically complete the schematic for all of circuit's states: Label the input(s) and the output(s), and connect the input(s) to Vcc or GND as needed to demonstrate the behavior of the circuit.

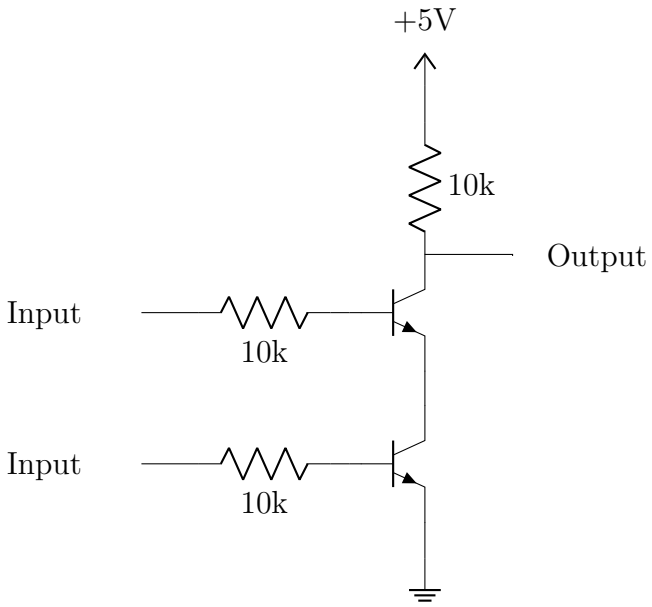
Note: You may use MS Paint or any drawing software to complete the schematic for all of the circuit's states.



Circuit: NAND gate.

Truth table:

Input A	Input B	Output Y
0	0	1
0	1	1
1	0	1
1	1	0



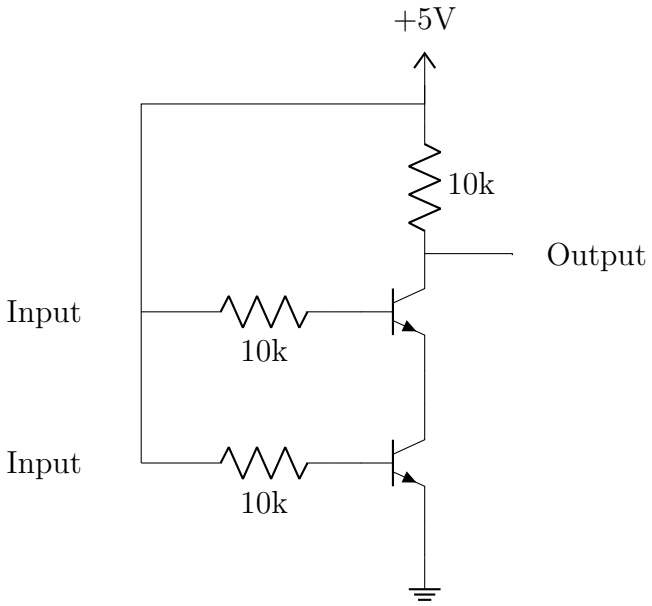


Figure 1: Both inputs HIGH, output LOW.

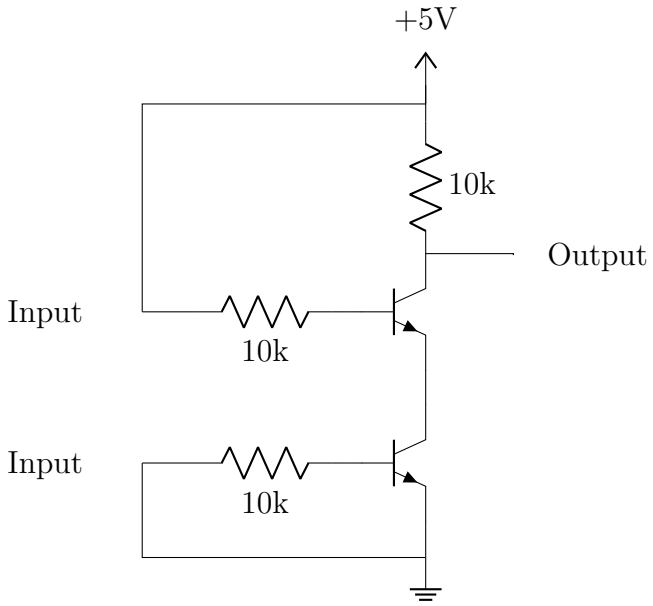


Figure 2: One input HIGH, one input LOW, output HIGH.

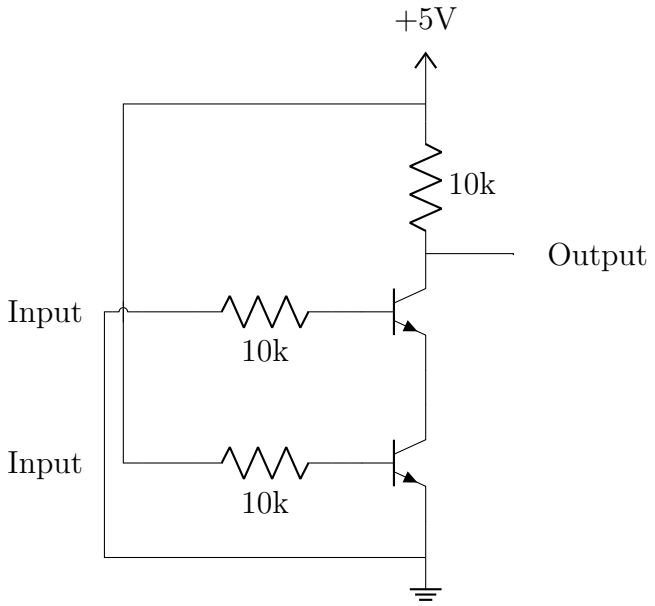


Figure 3: One input LOW, one input HIGH, output HIGH.

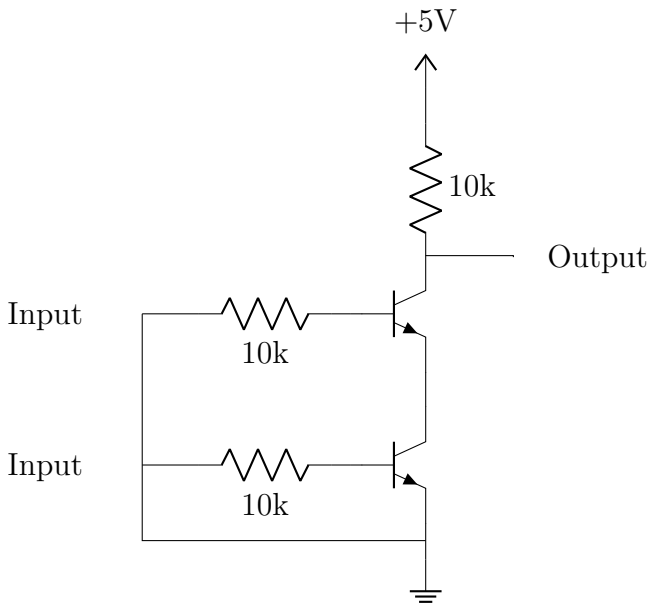


Figure 4: Both inputs LOW, output HIGH.

Question[25]: Design a binary multiplicator that multiplies three 4-bit numbers. Test your circuit on $1111_2 \times 1111_2 \times 1111_2$.

You have been provided with a 12-bit ripple carry adder subcircuit. It has inputs on one side and outputs on the other, such that two units can conveniently be connected to feed the output of one into the input of the other. Make good use of it.

Subcircuits are your friends and will make life easier.

Hints:

- Think the problem through, on paper even, before you start drawing the circuit. You will need to break down the problem into smaller subproblems.
- To provide ZERO or LOW to unused inputs, you can connect them to GND.

Solution:

Let the three numbers be A, B and C respectively. Let P be the product of A and B, and Q be the product of P and C. The circuit can be designed as follows:

								B_3	B_2	B_1	B_0
							$\times$	A_3	A_2	A_1	A_0
								A_0B_3	A_0B_2	A_0B_1	A_0B_0
						$+$	A_1B_3	A_1B_2	A_1B_1	A_1B_0	X
					$+$	A_2B_3	A_2B_2	A_2B_1	A_2B_0	X	X
				$+$	A_3B_3	A_3B_2	A_3B_1	A_3B_0	X	X	X
				P_7	P_6	P_5	P_4	P_3	P_2	P_1	P_0
							$\times$	C_3	C_2	C_1	C_0
				C_0P_7	C_0P_6	C_0P_5	C_0P_4	C_0P_3	C_0P_2	C_0P_1	C_0P_0
		$+$	C_1P_7	C_1P_6	C_1P_5	C_1P_4	C_1P_3	C_1P_2	C_1P_1	C_1P_0	X
	$+$	C_2P_7	C_2P_6	C_2P_5	C_2P_4	C_2P_3	C_2P_2	C_2P_1	C_2P_0	X	X
$+$	C_3P_7	C_3P_6	C_3P_5	C_3P_4	C_3P_3	C_3P_2	C_3P_1	C_3P_0	X	X	X
Q_{11}	Q_{10}	Q_9	Q_8	Q_7	Q_6	Q_5	Q_4	Q_3	Q_2	Q_1	Q_0